PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Variation in the estimated prevalence of multimorbidity: systematic review and meta-analysis of 193 international studies
AUTHORS	Ho, Iris Szu-Szu; Azcoaga-Lorenzo, Amaya; Akbari, Ashley; Davies, Jim; Hodgins, Peter; Khunti, Kamlesh; Kadam, Umesh; Lyons, Ronan; McCowan, Colin; Mercer, Stewart; Nirantharakumar, Krishnarajah; Guthrie, Bruce

VERSION 1 – REVIEW

REVIEWER	Violán, Concepción Institut Universitari d'Investigació en Atenció Primària Jordi Gol (IDIAP Jordi Gol)
REVIEW RETURNED	13-Oct-2021

GENERAL COMMENTS	The manuscript by Ho Iris Szu et al describes an exhaustive systematic Review regarding Multimorbidity prevalence studies published between database inception and 21 January 2020 considering the Prisma and Cochrane Database recommendations for Systematic Reviews. Although there are previous systematic reviews, this study has the advantage of including a very large number of studies. In addition, an effort has been made to avoid bias in the results, including multiple imputation. The results of the study are informative, but relevant to explain the prevalence of multimorbidity at the international level and by age group. Minor considerations in the discussion section.

REVIEWER	Roomaney, Rifqah South African Medical Research Council, Burden of Disease Research Unit, Tygerberg
REVIEW RETURNED	04-Jan-2022

GENERAL COMMENTS	General comments: Thank you for the opportunity to review this study. I believe that this is a well-conceived and important study. The methodology is clear. It provides estimates of multimorbidity that will be useful to other researchers. Beyond only the prevalence estimates, it also has important methodological findings regarding how variations in measurements of multimorbidity impact prevalence measurements. ABSTRACT: • The objective, as written, is vague. The reviewers could add a primary objective e.g. “to estimate the prevalence of multimorbidity in all age groups, globally” and a secondary objective e.g. “to

	analyse how measurement of multimorbidity impacts the estimated prevalence". • A limitation could be added for the exclusion of non-English studies. Introduction • Line 19: From my understanding, multimorbidity may be more common in people from low socioeconomic backgrounds - but only in certain settings. I'm not sure if this holds true for low and middle income countries. Can authors please check. • Line 43 – 46: Authors could make a stronger case for why their study is important, and what the implications are for better understanding variations in methods etc. Methods: • Line 4: I suggest authors remove "(n=566)". It does not seem relevant to this study and is reported in the referenced study. Results: • Page 11, Line 36: The 24 outlying studies have quite large sample sizes. What would have been the overall prevalence if these studies were included? Would it have been lower? • How many of the studies reported prevalence disaggregated by sex? Discussion: • Page 13, Line 3-4: Differences in search strategies could also account for a smaller number of LMIC studies included when compared to Nguyen? • Page 14, Line 5-6: Is there a recommendation around specific age-groups e.g. 5 or 10-year age bands? • Page 14, Line 16 – 22: I agree that using a common core set of conditions would be helpful in studies of multimorbidity, however, this must be carefully considered. For example, in low and middle income countries, it will be important to include HIV and other diseases important to those settings. In addition, many studies of multimorbidity (especially in LMICs) are based on secondary data analysis of national survey data which is limited in the disease conditions they include. • Page 14, Line 23 – 24: The finding that there was little difference in prevalence between self-report and administrative is surprising but also a powerful finding that will surely be cited in future studies. Table S4 • Is there information on the year that data was collected in each study? Not the date it was published. The year of data collected could reflect changes in the burden of disease. • Ref 145 – Please check the Weimann study prevalence. It is 2.7% not 27%. • Ref 170 & 171 – the Pati studies look like the same source of information.

REVIEWER	Svensson, Elisabeth Orebro University
REVIEW RETURNED	21-Jan-2022

GENERAL COMMENTS	Bmjopen-2021-057017 Variation in the estimated prevalence of multimorbidity: systematic review and meta-analysis of 194 international studies As a biostatistician and reviewer, I have been invited to review this manuscript with a particular emphasis on the statistical methods and analyses used. INTRODUCTION, RESEARCH QUESTION: This manuscript is part three of a comprehensive review regarding variations in measuring multimorbidity. However, the authors' published (ref 15) findings regarding definitions and measurements of multimorbidity are mentioned in the Methods section, but not in the Introduction/Background section. Does it mean that the results in this first study were not available when this manuscript were planned? However, the present manuscript, and especially the study aims and research questions, are based on the experiences from the first study – and should be part of the background (page 5 line 46) to the aim and the missing research questions of this manuscript. The study aim is described in general terms, referring to variations in prevalence of multimorbidity, in total, and in subgroups, but also regarding "associations with study and multimorbidity measure characteristics". Please note that the aim should be the link between the research problem as presented in the Introduction/Background, the design of the data collection for this review, and to the statistical methods. The solved problem will then be presented in the Result, and commented in the Discussion. Please, specify the research questions of this review study. METHODS page 5: Data collection- page 7: Risk of bias It is good that the definition of multi-morbidity used is mentioned. Major parts of these sections are almost identical to the published parts of the first parts of this project (ref. 15). Data analysis pages 7-9: The content of this section should be the link from the research question and to the data collection for this study, and to the statistical methods of descriptions and evaluations. Since this important part of a scientific study is missing, the data analysis plan needs a comprehensive revision. Some examples: Page 7, line 43-: This sentence is non-informative. Statistical methods for describing the data-sets/groups of interest, and the variables must be specified. For example, a measure of central tendency, and the variability of the data: The mean (SD), and range (min, max), are appropriate for quantitative continuous, symmetric, data, and Median (Q1, Q3) and range (min. max) is appropriate for quantitative, continuous skewed data, and for count data, and for qualitative ordered data.

	Please note that the median approach should be described Md (Q1, Q3). The concept interquartile range, IQR, defines the numerical difference between the third and first quartile in quantitative data. The remaining information on the pages 7-9: Data analysis is a mixture of analysis plans and results – but unrelated to research questions – and give rise to many questions; why, how, which question, which variable, problem, models???? It seems that these pages contain results from the previous study, and also from this study: “missing values, etc.” together with a kind on data analysis plan for this systematic review. Why imputation, why try to homogenize other studies, why manipulation of the reviewed papers? What is the meaning of a systematic review after manipulation of the results of the papers? Also, please note that the aim of a data analysis (in this case-page 9, line 5) is never to “calculate a p-value”! RESULTS: Because of the lack of links from research questions and variables, to methods, and to analyses plans, the results cannot be reviewed. Some common rules: Never use subjective interpretations in the results section. Subjective expressions can be used in the discussion section. Quantitative data could be described by the mean approach provided symmetric distributions. The mean and SD define, and describe, the symmetric probability distribution of quantitative continuous data called the Normal (or Gauss) distribution. It is well-known that 95% of normally distributed data will be found within the range of (mean value + 1.96 SD), and it is extremely unlikely to find a normally distributed observation outside the range of (mean + 3.9 SD). For example: Table 1: Does the variable “mean age” in the table means that the mean age of each paper’s study group is the basis for the mean of the mean ages? Then the study-mean ages range from 32.3 to 83.8 years. Both the mean- and the median approach is used in the Table 1 for description. However, the authors should consider, that the distribution of the mean ages are skewed, which invalidates the use of the mean approach. A quick check of the range of (mean value + 1.96 SD) reveals that 95% of the mean ages will range from 36.0 to 86.2 years, which is unlikely (>83.8 years)! The range reveals the skewed distribution, which means that the median approach must be used for statistical description: 62.4 (Q1 49.9, Q3 72.3) of study mean ages. Please check the statistical description of the highly skewed variable: number of conditions. What is meant by the calculations of mean and median of “prevalence of multimorbidity”?

	Reconsider these statistical descriptions. ---------- Because of the lack of research questions, as well as lack of specified data analysis plans to each question, the results are not interpretable. A large number of hypothesis tests have been used. There is a slight impression of p-value “fishing”. Please remember that the aim of a hypothesis test is to calculate the probability of obtaining the observed, or more extreme data, in samples of independent data when the null hypothesis is true. The p-value is the probability of obtaining the observed result or a more unlikely one, when the null hypothesis is true. Also, please note that a so-called non-significant p-value, (see e.g. page 11) never prove a null-hypothesis, but only that there is not enough evidence to reject a null-hypothesis in favour of the hypothesis of a significant result (difference, relationship, etc). Ref: Altman DG (1991) Practical statistics for medical research. London: Chapman & Hall, pp 165-175, 477-500. Altman D. Statistical reviewing for medical journals. Statistics in Medicine 1998; 17, 2661— 2674.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1:		
5. The manuscript by Ho Iris Szu et al describes an exhaustive systematic Review regarding Multimorbidity prevalence studies published between database inception and 21 January 2020 considering the Prisma and Cochrane Database recommendations for Systematic Reviews. Although there are previous systematic reviews, this study has the advantage of including a very large number of studies. In addition, an effort has been made to avoid bias in the results, including multiple imputation.		

The results of the study are informative, but relevant to explain the prevalence of multimorbidity at the international level and by age group. Minor considerations in the discussion section.		

6. Discussion I would suggest adding this quote, it reinforces the conclusions reached in your article, As I am one of the authors, please do not feel obligated Prevalence, Determinants and Patterns of Multimorbidity in Primary Care: A Systematic Review of Observational Studies. PLoS ONE 9(7): e102149. doi:10.1371/journal.pone.0102149	This study has now been included.	Page 12 and Reference 34
7. Regardin the USA. I would suggest adding another cause. Another reason may be due to the financing of the said system. The USA has a private system based on specialists, is not a system based on primary care, this increases the number of diagnoses when consulting multiple professionals.	Revised— “A possible explanation for the higher prevalence in North America is that private or insurance-based healthcare systems are more likely to code conditions since it affects remuneration, as well as cultural differences in relation to over-diagnosis and medicalisation [42]. 42. Kale, M.S. and D. Korenstein, Overdiagnosis in primary care: framing the problem and finding solutions. BMJ, 2018. 362(k2820): p. 1-48	Page 12
Reviewer 2:		

8. General comments: [redacted] Thank you for the opportunity to review this study. I believe that this is a well-conceived and important study. The methodology is clear. It provides estimates of multimorbidity that will be useful to other researchers. Beyond only the prevalence estimates, it also has important methodological findings regarding how variations in measurements of multimorbidity impact prevalence measurements.		
9. ABSTRACT: [redacted] The objective, as written, is vague. The reviewers could add a primary objective e.g. “to estimate the prevalence of multimorbidity in all age groups, globally” and a secondary objective e.g. “to analyse how measurement of multimorbidity impacts the estimated prevalence”.	Revised— Abstrac t Objective: (1) To estimate the pooled prevalence of multimorbidity in all age groups, globally. (2) To examine how measurement of multimorbidity impacted the estimated prevalence. Introduction [redacted] “This review aimed to examine the pooled prevalence of multimorbidity in all age groups, globally and how measurement of multimorbidity impacted the estimated prevalence.”	Page 2 and 4
10. A limitation could be added for the exclusion of non-English studies.	Added— “The exclusion of non-English studies in this review may also limit the generalisability of the research findings.”	Page 13
11. Introduction: [redacted] Line 19: From my understanding, multimorbidity may be more common in	Revised as below:	Page 4

people from low socioeconomic backgrounds - but only in certain settings.		

I'm not sure if this holds true for low and middle income countries. Can authors please check.	"Multimorbidity affects all groups of society, but is known to be more common in older people, in women, and in those from low socio-economic backgrounds particularly in high- income countries [5-7]. In low- and middle-income countries, people living in urban areas, on the other hand, were found to have a higher rate of multimorbidity [8]." 8. Asogwa, O.A., et al., Multimorbidity of non-communicable diseases in low-income and middle-income countries: a systematic review and meta-analysis. BMJ Open, 2022. 12(1): p. 1-16.	
12. Line 43 – 46: Authors could make a stronger case for why their study is important, and what the implications are for better understanding variations in methods etc.	Added— "Much of the research up to now has not quantitatively investigated the variation in multimorbidity prevalence and its influencing factors in much detail. Understanding the links between prevalence estimates and measurement approaches can better inform and support future development of multimorbidity measurement guidelines."	Page 4
13. Methods: Line 4: I suggest authors remove "(n=566)". It does not seem relevant to this study and is reported in the referenced study.	Removed	
14. Results:	We have now added the information in Table S2. The pooled prevalence for the 217 studies	Table S2

Page 11, Line 36: The 24 outlying studies have quite large sample sizes. What would have been the overall prevalence if these studies were included? Would it have been lower?	is 41.1% (95%CI=37.7-44.6), which is very similar to the primary analysis pooled prevalence with 193 studies (42.4% 95%CI=38.9-46.0).	(supplementary)
15. How many of the studies reported prevalence disaggregated by sex?	This was not commonly reported and was not extracted. We therefore do not have the data available to quantify it.	
16. Discussion: Page 13, Line 3-4: Differences in search strategies could also account for a smaller number of LMIC studies included when compared to Nguyen?	The number of studies from both high-income countries and LMIC included in this review was higher than Nguyen's. To avoid confusion, we revised the phrase as below— "Nguyen et al. in their review showed a statistically significantly higher pooled prevalence in high-income countries (the pooled prevalence from 18 studies was 37% compared to 36.8% in this review of 145 studies) than low or middle-income countries (the pooled prevalence from 31 studies was 29% compared to 44.3% in this review of 48 studies). This difference in findings may be due to the inclusion in our review of a larger number of studies from high-income or upper middle-income countries." We removed this phrase "whereas very few studies were from low-income or lower middle-income countries"	Page 12
17. Page 14, Line 5-6: Is there a recommendation around specific age- groups e.g. 5 or 10-year age bands?	Added— "We therefore strongly recommend that as well as overall prevalence, future studies should clearly report multimorbidity prevalence stratified by age, in 5-year age	Page 13

	bands to ensure granularity, and by sex at a minimum, and ideally by ethnicity and socio- economic status."	
18. Page 14, Line 16 – 22: I agree that using a common core set of conditions would be helpful in studies of multimorbidity, however, this must be carefully considered. For example, in low and middle income	In the recommendation section, we described the importance of having a core condition set as well as bespoke conditions chosen for the context and purpose. —"Parallel reporting of the bespoke set chosen for the context and purpose, and a core set would improve	Page 13

countries, it will be important to include HIV and other diseases important to those settings. In addition, many studies of multimorbidity (especially in LMICs) are based on secondary data analysis of national survey data which is limited in the disease conditions they include.	comparability of prevalence estimates, and help identify the additional value of any bespoke multimorbidity measures."	
19. Page 14, Line 23 – 24: The finding that there was little difference in prevalence between self-report and administrative is surprising but also a powerful finding that will surely be cited in future studies.	We also thought this new finding provides important implications.	
20. Table S4 Is there information on the year that data was collected in each study? Not the date it was published. The year of data collected could reflect changes in the burden of disease.	Unfortunately, this piece of information was not collected, partly because (unlike year of publication) it is not always clearly stated in papers, and some studies collected data over several years.	
21. Ref 145 – Please check the Weimann study prevalence. It is 2.7% not	Thank you for identifying this error, which we have corrected and have re-run all of the	Page11-13,

27%.	analyses. All results and graphs have been updated in the main manuscript and supplementary document. The main difference found is that the estimated prevalence of multimorbidity in Africa is now statistically significantly lower than in other continents. Added— 1. Africa (OR 0.3 95%CI 0.1-0.6) in results (page 11) 2. African studies had the lowest estimated prevalence than other continents. (page 12) 3. On the other hand, the lower estimated multimorbidity prevalence in African studies could be attributed to the predominance of infectious diseases and inadequate access to medical care including diagnostic services [43]. (page 12-13) Sensitivity analysis is now similar to primary analysis for the “study setting” variable. The pooled prevalence in both primary and sensitivity analyses was significantly higher in primary care settings compared to community settings. Thus, we removed the phrase— “In respect to study settings, the pooled prevalence in sensitivity analysis was statistically significantly higher in primary care compared to community in both unadjusted and adjusted models, whereas in primary analysis the difference was not statistically significant after controlling for study and measure characteristics.” (page 11)	Table 2

22. Ref 170 & 171 – the Pati studies look like the same source of information.	We have now removed Ref 171 (Pati et al., 2017) but keep Ref 170 (Pati et al., 2019), as Ref 170 provided data on the total sample size and the number of people with MM, so we could calculate raw MM prevalence. All results and graphs have been updated as indicated above.	
Reviewer 3:		
23. The authors' published (ref 15) findings regarding definitions and measurements of multimorbidity are mentioned in the Methods section, but	We have now cited the first review in the introduction.	Page 4-5

not in the Introduction/Background section. Does it mean that the results in this first study were not available when this manuscript were planned? However, the present manuscript, and especially the study aims and research questions, are based on the experiences from the first study – and should be part of the background (page 5 line 46) to the aim and the missing research questions of this manuscript. The study aim is described in general terms, referring to variations in prevalence of multimorbidity, in total, and in subgroups, but also regarding "associations with study and multimorbidity measure characteristics".	"Previous systematic reviews have identified issues in the measurement of multimorbidity, related to the choice of chronic conditions counted in measures, the categorisation of conditions and diseases, and the counting or weighting method used [11-13]. Although weighted measures are often used when the purpose of measurement is to predict future outcomes, a simple count of conditions remains the most commonly-used method for the measurement of multimorbidity, and is optimal for estimating multimorbidity prevalence [13, 14]. However, the estimated prevalence of multimorbidity varies widely in the literature ranging from 3.5% to 100% [15], likely reflecting a combination of varying measures and varying populations studied [16]. Much of the research up to now has not	

Please note that the aim should be the link between the research problem as presented in the Introduction/Background, the design of the data collection for this review, and to the statistical methods. The solved problem will then be presented in the Result, and commented in the Discussion. Please, specify the research questions of this review study.	quantitatively investigated the variation in multimorbidity prevalence and its influencing factors in much detail. Understanding the links between prevalence estimates and measurement approaches can better inform and support future development of multimorbidity measurement guidelines. Therefore, this review aimed to examine the pooled prevalence of multimorbidity in all age groups globally and how measurement of multimorbidity impacted the estimated prevalence." Aside from research aims addressed in the introduction, we have now added research questions as well. Research questions • What is the pooled prevalence of multimorbidity and does it differ between different age groups? • What are the factors that influenced the variation in prevalence estimates across studies?	
24. METHODS page 5: Data collection- page 7: Risk of bias It is good that the definition of multi-morbidity used is mentioned. Major parts of these sections are almost identical to the published parts of the first parts of this project (ref. 15).	The overarching focus for the two reviews is multimorbidity measurement but with different research questions (as detailed in the registered protocol), and hence the methods will have considerable overlap. However, the findings are non-overlapping.	
25. Page 7, line 43-: This sentence is non-informative. Statistical methods for describing the data-sets/groups of interest, and the	Detailed information has now been provided.	Page 7

variables must be specified. For example, a measure of central tendency, and the variability of the data: The mean (SD), and range (min, max), are appropriate for quantitative continuous, symmetric, data, and Median (Q1, Q3) and range (min. max) is appropriate for quantitative, continuous skewed data, and for count data, and for qualitative ordered data.	"Descriptive statistics were used to summarise study characteristics. Since distributions were skewed, median and interquartile range were used to measure the central tendency and examine variability of variables such as mean age and number of conditions. Categorical (e.g. continent, study population, and data source) and ordinal data (e.g. country income and risk of bias) were examined using frequency tables." We have now removed IQR and changed to Median (Q1,Q3)	Table 1

Please note that the median approach should be described Md (Q1, Q3). The concept interquartile range, IQR, defines the numerical difference between the third and first quartile in quantitative data.		
26. The remaining information on the pages 7-9: **Data analysis** is a mixture of analysis plans and results – but unrelated to research questions – and give rise to many questions; why, how, which question, which variable, problem, models???? Because of the lack of links from research questions and variables, to methods, and to analyses plans, the results cannot be reviewed.	The analytical plan is related to the research questions. This is a meta-analysis and meta- regression study. Before carrying out meta-analysis, we firstly performed data exploration (understanding data distribution and associations between predictors). Then we pooled the prevalence of multimorbidity across studies and found high heterogeneity in prevalence estimates. Due to the high level of heterogeneity, data pre-processing and management were necessary to minimise bias caused by outliers and missing data. We used	

	recommended statistical approaches in meta-analysis to identify outlying studies (e.g.		
It seems that these pages contain results from the previous study, and also	studentised residuals, leave-one-out analysis), and used multiple imputation to impute		
from this study: “missing values, etc.” together with a kind on data analysis	missing data. In this review, all included study reported a prevalence of multimorbidity.		
plan for this systematic review. Why imputation, why try to homogenize	There is therefore no missing data for the outcome, and the pooled estimate is derived from		
other studies, why manipulation of the reviewed papers?	meta-analysis for complete data. There is no manipulation. In the meta-regression there are		
What is the meaning of a systematic review after manipulation of the results	missing data in the mean age and number of conditions variables. We imputed this missing		
of the papers?	data as recommended by prior research. We have now added the evidence—“Previous	Page 8	
	research has shown that complete case removal (removing missing data in a data set) in		
	meta-regression could lead to biased coefficient estimates of predictors (varied widely from		
	complete-data estimates), whereas multiple imputation was found to perform well at		
	generating estimates that were close to complete-data estimates [25]. Therefore, in this		
	review, multiple imputation with 60 imputed datasets and 10 iterations was conducted.”		
	25. Ellington, E.H., et al., Using multiple imputation to estimate missing data in meta-		
	regression. Methods in Ecology and Evolution, 2015. 6(2): p. 153-163.		
	These analytical approaches were conducted with rigour and transparency in an attempt to		
	minimise bias. To evaluate the performance of these analytical approaches, we did		
	sensitivity analysis to see if there is a significant difference in results (most of which were		

	similar) and measured Fraction of Missing Information (FMI) to identify level of uncertainty caused by multiple imputation (the FMI for all variables is smaller than 0.3). We did not know what the results would be like before conducting outlier tests and multiple imputation. Any differences found following sensitivity analysis were reported in the paper. Our transparent reporting of the analytical plan and results should be able to reflect our rigour in application. We accept that there is a sense that some of the description of this could be presented as 'results' but we believe it makes more sense to include details of this pre-processing in methods since it may distract from the core findings in results (and this is a common	

	strategy in articles primarily intended for a clinical or general readership). If you think that they should be moved to the start of findings, then please let us know. We do not understand the comment that "it seems these pages contain results from the previous study". The findings relate to different pre-specified objectives in the PROSPERO registered protocol (the other paper addresses the first two of these; this paper addresses the third). The findings are therefore quite distinct.	
27. Also, please note that the aim of a data analysis (in this case- page 9, line 5) is never to "calculate a p-value"! Because of the lack of research questions, as well as lack of specified data	We did not manipulate the p values. Permutation test is a resampling method that has been recommended in meta-regression. This method is used to validate the final model by shuffling the order of the data, and re-calculating p values to check if there are changes in	Page 8

analysis plans to each question, the results are not interpretable. A large number of hypothesis tests have been used. There is a slight impression of p-value“fishing”. Please remember that the aim of a hypothesis test is to calculate the probability of obtaining the observed, or more extreme data, in samples of independent data when the null hypothesis is true. The p-value is the probability of obtaining the observed result or a more unlikely one, when the null hypothesis is true. Also, please note that a so-called non-significant p-value, (see e.g. page 11) never prove a null-hypothesis, but only that there is not enough evidence to reject a null-hypothesis in favour of the hypothesis of a significant result (difference, relationship, etc).	p-values following shuffling the order of the data. We agree that the original phrase might have caused confusion — “A permutation test with 1000 permuted datasets was performed on the final meta-regression model to calculate p value and avoid type 1 error [30]”. We have now revised it— “A permutation test with 1000 permuted datasets was conducted to validate the robustness of the final model by rearranging and shuffling the order of the data and re-calculating p-values to check whether there is type 1 error [32].” 32. Gagnier, J.J., et al., An empirical study using permutation-based resampling in meta-regression. Systematic reviews, 2012. 1(1): p. 1-9.	
28. Some common rules: Never use subjective interpretations in the results section. Subjective expressions can be used in the discussion section.	All the subjective terms (“we”) were removed from the result section now.	
29. Quantitative data could be described by the mean approach provided	Since the continuous variables have skewed distributions, we have amended table 1 as	Table 1

symmetric distributions. The mean and SD define, and describe, the symmetric probability distribution of quantitative continuous data called the Normal (or Gauss) distribution. It is well-known that 95% of normally distributed data will be found within the range of (mean value + 1.96 SD), and it is extremely unlikely to find a normally distributed observation outside the range of (mean + 3.9 SD). Does the variable “mean age” in the table means that the mean age of each paper’s study group is the basis for the mean of the mean ages? Then the study-mean ages range from 32.3 to 83.8 years.Both the mean- and the	suggested to only show median (Q1, Q3) and range. Yes, what is reported is the median of “mean age of the study population”. We have amended the table to make this clearer.	

VERSION 2 – REVIEW

REVIEWER	Violán, Concepción Institut Universitari d'Investigació en Atenció Primària Jordi Gol (IDIAP Jordi Gol)
REVIEW RETURNED	23-Mar-2022

GENERAL COMMENTS	The authors have applied the changes suggested by the reviewers, very good paper. NO additional requirement

REVIEWER	Roomaney, Rifqah South African Medical Research Council, Burden of Disease Research Unit, Tygerberg
REVIEW RETURNED	24-Mar-2022

GENERAL COMMENTS	All queries were resolved in the revision